

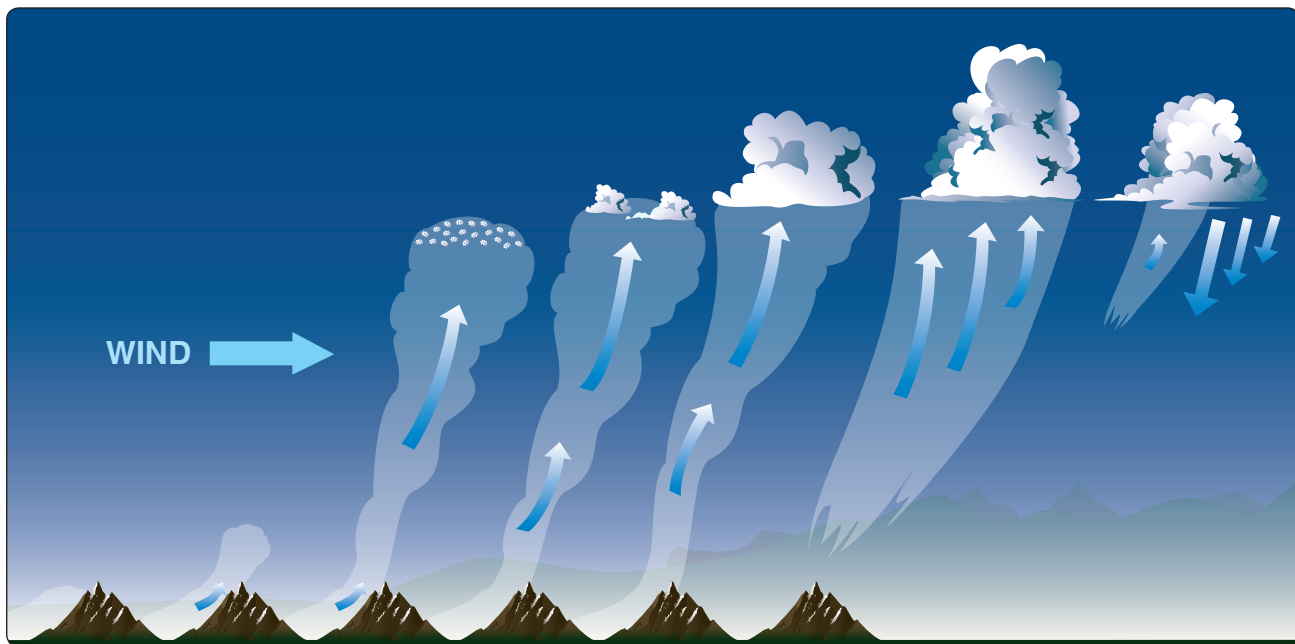


Figure 9-10. Lifecycle of a typical thermal with cumulus cloud.

The DALR is the rate at which the temperature of unsaturated air changes as a parcel ascends or descends through the atmosphere which is approximately 9.8°C per 1 kilometer. On a thermodynamic chart, parcels cooling at the DALR are said to follow a dry adiabatic. A parcel warms at the DALR as it descends. In reality, heat transfer often occurs. For instance, as a thermal rises, the circulation in the thermal itself (recall the bubble model) mixes in surrounding air. Nonetheless, the DALR is a good approximation.

The DALR represents the lapse rate of the atmosphere when it is neutrally stable. If the ambient lapse rate in some layer of air is less than the DALR (for instance, 1°C per 1,000 feet), then that layer is stable. If the lapse rate is greater than the DALR, it is unstable. An unstable lapse rate usually occurs within a few hundred feet of the heated ground. When an unstable layer develops aloft, the air quickly mixes and reduces the lapse rate back to DALR. It is important to note that the DALR is not the same as the standard atmospheric lapse rate of 2°C per 1,000 feet. The standard atmosphere is a stable one.

Another way to understand stability is to imagine two scenarios, each with a different temperature at 3,000 feet above ground level (AGL), but the same temperature at the surface, nominally 20°C . In both scenarios, a parcel of air that started at 20°C at the surface has cooled to 11°C by the time it has risen to 3,000 feet at the DALR. In the first scenario, the parcel is still warmer than the surrounding air, so it is unstable and the parcel keeps rising—a good thermal day. In the second scenario, the parcel is cooler than the surrounding air, so it is stable and sinks. The parcel in the second scenario would need to be forced to 3,000 feet AGL

by a mechanism other than convection, such as being lifted up a mountainside or a front. [Figure 9-11]

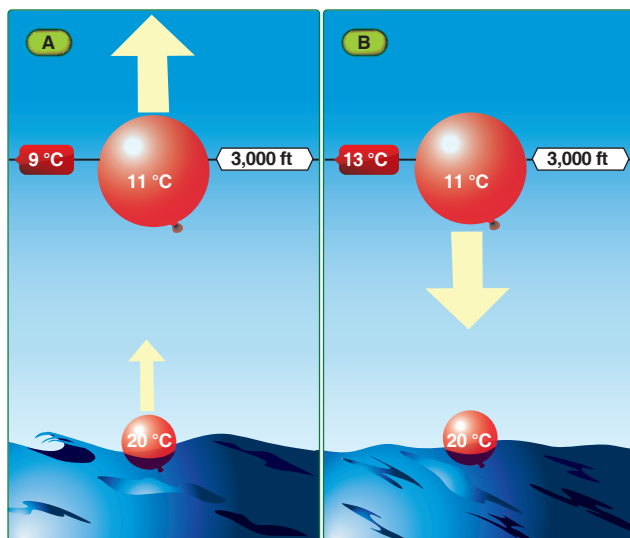


Figure 9-11. Unstable (A) and stable (B) parcels of air.

Figure 9-11 also illustrates factors leading to instability. A stable atmosphere can turn unstable in one of two ways. First, if the surface parcel warms by more than 2°C (to greater than 22°C), the layer to 3,000 feet then becomes unstable in the second scenario. Thus, if the temperature of the air aloft remains the same, warming the lower layers causes instability and better thermal soaring. Second, if the air at 3,000 feet is cooler, as in the first scenario, the layer becomes unstable. Thus, if the temperature on the ground remains the same, cooling aloft causes instability and better thermal soaring. If the temperatures aloft and at the surface warm or cool